WORKED SOLUTION

STEP I 2010, Question 1

Pure • Completing the Square

STEP I 2010, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q1

The key idea

The identity rewrites an awkward quadratic as a sum of two squares plus a constant. Once you notice that the second equation can be written using the *same* two brackets, $x-y+2$ and $2x-y$, the problem becomes two simple equations in two new unknowns. These are linear in the squares of the new unknowns, so we can solve them directly.

Solution

The identity

Expanding the right-hand side,

$$\begin{aligned} &a(x-y+2)^2+b(cx+y)^2+d \\ &\quad = a(x^2+y^2-2xy+4x-4y+4)+b(c^2x^2+2cxy+y^2)+d \\ &\quad = (a+bc^2)x^2+(a+b)y^2+(2bc-2a)xy+4ax-4ay+(4a+d). \end{aligned}$$

Since this is an identity, we can compare coefficients with $5x^2+2y^2-6xy+4x-4y$:

$$\begin{aligned} x: &\quad 4a=4, && \text{so } a=1; \\ y: &\quad -4a=-4, && \text{which agrees;} \\ y^2: &\quad a+b=2, && \text{so } b=1; \\ xy: &\quad 2bc-2a=-6, && \text{so } 2c=-4,\ c=-2; \\ x^2: &\quad a+bc^2=5, && \text{check: } 1+4=5; \\ \text{constant}: &\quad 4a+d=0, && \text{so } d=-4. \end{aligned}$$

Note that the x^2 coefficient on its own only gives $c^2=4$; it is the xy coefficient that fixes the sign. So

$$a=1, \qquad b=1, \qquad c=-2, \qquad d=-4,$$

that is,

$$5x^2+2y^2-6xy+4x-4y \equiv (x-y+2)^2+(2x-y)^2-4,$$

where we have written $(-2x+y)^2$ as $(2x-y)^2$.

The simultaneous equations

By the identity, the first equation becomes

$$(x-y+2)^2+(2x-y)^2=13.$$

Ask yourself The second equation looks different. Can it be written using the same two brackets?

Its terms in x and y are $8x-8y$, twice those of the first equation. This suggests trying $2(x-y+2)^2$, which has the right linear terms:

$$2(x-y+2)^2=2x^2+2y^2-4xy+8x-8y+8.$$

The rest of the left-hand side is

$$6x^2+3y^2-8xy+8x-8y-2(x-y+2)^2=4x^2+y^2-4xy-8=(2x-y)^2-8.$$

So the second equation is $2(x-y+2)^2+(2x-y)^2-8=14$, or

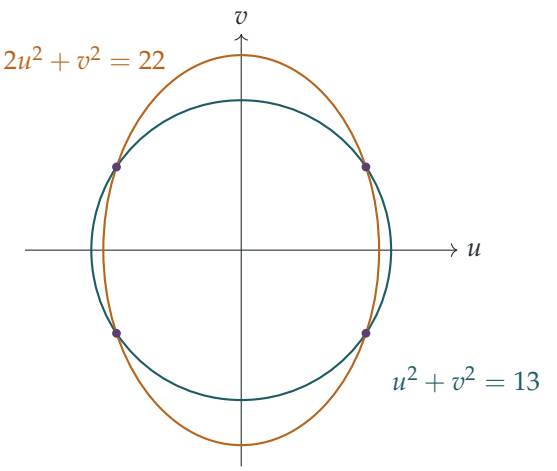
$$2(x-y+2)^2+(2x-y)^2=22.$$

(You could also find this the way we found the identity, by comparing coefficients in $\alpha(x-y+2)^2+\beta(\gamma x+y)^2+\delta$.)

Now let $u=x-y+2$ and $v=2x-y$. The system becomes

$$u^2+v^2=13, \qquad 2u^2+v^2=22.$$

Subtracting gives $u^2=9$, and then $v^2=4$. So $u=\pm3$ and $v=\pm2$, with the signs chosen independently, which gives four pairs (u,v).



To get back to x and y, subtract: $v-(u-2)=(2x-y)-(x-y)=x$, so

$$x=v-u+2, \qquad y=2x-v=v-2u+4.$$

Each (u,v) gives exactly one (x,y), because this change of variables is linear and invertible. The four solutions are:

u	v	x	y
3	2	1	0
3	-2	-3	-4
-3	2	7	12
-3	-2	3	8

So

$$(x,y)=(1,0), \quad (-3,-4), \quad (7,12), \quad (3,8).$$

Checks

Substitute $(1,0)$: the first equation gives $5+4=9$ and the second $6+8=14$. For $(7,12)$: $245+288-504+28-48=9$ and $294+432-672+56-96=14$. A computer confirmed the two identities at a thousand random rational points and confirmed that all four pairs satisfy both equations. A brute-force search over all integer pairs with $|x|,|y|\leq 50$ found exactly these four solutions. Two conics meet in at most four points, so we cannot have missed any.

Common slips

- Taking $c=2$ from $c^2=4$ without checking the xy coefficient.
- Keeping only the positive square roots, and so finding one solution instead of four.
- Eliminating a variable directly from the original equations. This gives a quartic, which is very hard work by hand.

Beyond STEP

Writing a quadratic form as a combination of squares is the start of the theory of quadratic forms. By diagonalising a symmetric matrix you can always find a linear change of variables that turns $Ax^2+Bxy+Cy^2$ into $\lambda_1u^2+\lambda_2v^2$, and this is how conics are classified. Here both equations happened to share the same pair of squares, which is why they decoupled so neatly. Strang's *Introduction to Linear Algebra* covers symmetric matrices and quadratic forms clearly.

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